

BLOOD SMEAR DIAGNOSTIC SUMMARY

Patient: Test Patient

ID: PH-12345

Date: 2025-12-22 19:12

■ PARASITIC INFECTIONS

Test	Result	Finding	Clinical Interpretation
Malaria (P. falciparum/vivax)	✓ NOT DETECTED	0 structures detected	No parasite-like structures identified in analyzed regions.
Method	YOLO Object Detection	Stages: None	Negative screening. Clinical suspicion requires confirmation with microscopy.

■ HEMOGLOBINOPATHIES (Sickle Cell & Thalassemia)

Condition	Result	Findings	Interpretation
Sickle Cell	■ TRACE	6 cells (2.3%)	Trace sickle cells (2.3%). May indicate carrier status.
Thalassemia	■ DETECTED	93 indicators (36.3%)	Significant thalassemia indicators (36.3%): target cells, teardrops, microcytes.
Next Steps			Sickle: Clinical correlation advised Thalassemia: Hemoglobin electrophoresis + CBC with RBC indices recommended

■ ANEMIA SCREENING (RBC Morphology)

Anemia Type	Indicator	Findings	Interpretation
Iron Deficiency (Microcytic)	■ BORDERLINE	15.2% microcytes	Mild microcytosis. Iron studies recommended.
Size Variation (Anisocytosis)	✓ NORMAL	Size CV: 18.5%	Normal size variation.

■ INFECTION INDICATORS (WBC Differential Count)

Cell Type	Count	%	Normal Range	Clinical Significance
Neutrophil	1	50.0%	40-70%	✓
Lymphocyte	0	0.0%	20-40%	Immunodeficiency, steroid use
Monocyte	0	0.0%	2-10%	✓
Eosinophil	1	50.0%	1-5%	Parasites, allergies, asthma
Basophil	0	0.0%	0-2%	✓

Clinical Interpretation: ↓ Lymphocyte (0.0%): Immunodeficiency, steroid use; ↓ Monocyte (0.0%): Bone marrow disorder; ↑ Eosinophil (50.0%): Parasites, allergies, asthma

■ LEUKEMIA / MALIGNANCY SCREENING

Type	Result	Findings	Clinical Action
ALL (Acute Lympho.)	✓ NEGATIVE	Lymphocytes: 0.0%	No lymphocytic predominance.
CML (Chronic Myeloid)	✓ NEGATIVE	Neutrophils: 50.0%	No myeloid predominance.

■ **IMPORTANT:** This system classifies 5 normal WBC types only. Atypical cells, blasts, or immature forms cannot be detected and require manual review by hematologist.

DISCLAIMER: This is an AI-assisted screening report for research and educational purposes only. NOT FOR DIAGNOSTIC USE. All findings require confirmation by trained medical professionals using standard laboratory methods (thick/thin smear microscopy, CBC, hemoglobin electrophoresis, flow cytometry).

Methods: YOLO11n object detection (malaria), ResNet34 classification (WBC, RBC), Instance segmentation (RBC morphology). Generated: 2025-12-22 19:12:23